

Proyek ISYS6895003  
Sistem Informasi <Retail Fashion Data>

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Kelompok : 07

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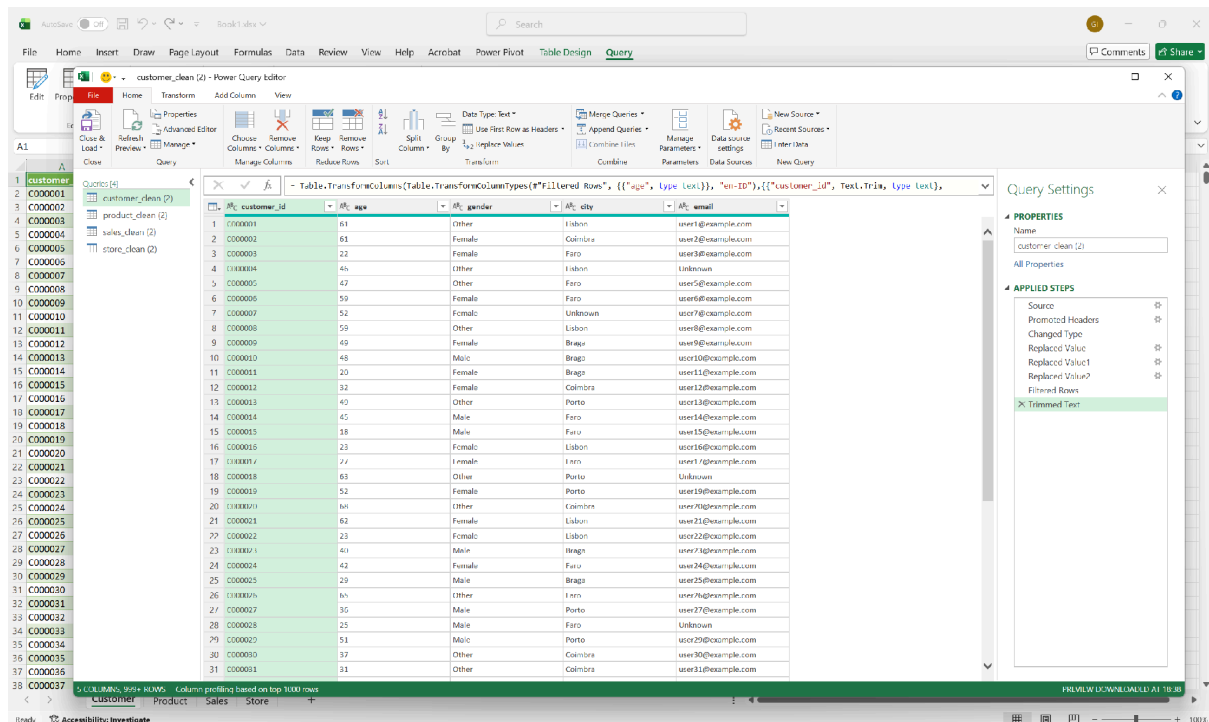
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**BINUS UNIVERSITY**  
**Jakarta**  
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## 1.Data Cleaning and Preparation

- Manipulating Text



Pada tabel customer, kami melakukan Replaced Value pada kolom gender, dilakukan cleaning nilai pada kolom gender menggunakan fitur Transform > *Replace Values* di Power Query. Hal ini diperlukan karena ditemukan inkonsistensi penulisan yang merujuk pada kategori yang sama, seperti "Male" dan "male" yang keduanya seharusnya direpresentasikan sebagai nilai tunggal "M", serta "FEMALE" dan "Female" yang distandarisasi menjadi "F". Selain itu, nilai tidak valid seperti "???" yang tidak merepresentasikan kategori gender yang dikenali diganti dengan nilai "Unknown" agar data tetap dapat dianalisis tanpa menghilangkan baris tersebut dari dataset. Proses Replace Values ini memastikan kolom gender hanya mengandung nilai yang konsisten dan terstandarisasi, sehingga analisis berdasarkan segmentasi gender dapat dilakukan secara akurat.

Proses pembersihan data diawali dengan menghapus spasi yang tidak diperlukan pada seluruh kolom bertipe teks menggunakan fitur Transform > *Trim* di Power Query. Langkah ini dilakukan untuk memastikan tidak ada karakter spasi tersembunyi di awal maupun di akhir nilai data yang dapat menyebabkan inkonsistensi saat proses analisis, penggabungan tabel, maupun pemfilteran data. Dengan menerapkan Trim secara menyeluruh pada semua kolom teks, seluruh nilai data menjadi seragam dan bersih dari karakter yang tidak terlihat namun berpotensi mengganggu proses pengolahan data selanjutnya.

Formula Bar: `= Table.ReplaceValue(@"Trimmed Text", "!!!", "Unknown", Replacer.ReplaceText, {"category"})`

| product_id | category    | color   | size | season | supplier  | cost_price |
|------------|-------------|---------|------|--------|-----------|------------|
| P000001    | Bottoms     | Black   | XL   | Spring | suppl/erb | 7826       |
| P000002    | Unknown     | Black   | M    | Summer | suppl/ner | 196.7      |
| P000003    | Tops        | Black   | XL   | Summer | suppl/ner | 2762       |
| P000004    | Accessories | White   | XL   | Winter | suppl/erb | 743        |
| P000005    | Accessories | Unknown | XL   | Fall   | suppl/cra | 1204       |
| P000006    | Shoes       | Black   | M    | Fall   | suppl/erc | 6869       |
| P000007    | Tops        | Blue    | S    | Fall   | suppl/erl | 1353       |
| P000008    | Unknown     | Yellow  | M    | Fall   | suppl/erc | 5003       |
| P000009    | Tops        | Yellow  | XS   | Summer | suppl/erh | 1411       |
| P000010    | Accessories | Black   | S    | Winter | suppl/erd | 1968       |
| P000011    | Bottoms     | Green   | XS   | Summer | suppl/crd | 4184       |
| P000012    | Tops        | Green   | XL   | Fall   | suppl/erh | 975        |
| P000013    | Accessories | White   | S    | Fall   | suppl/ers | 591        |
| P000014    | Shoes       | Black   | XL   | Fall   | suppl/crd | 3507       |
| P000015    | Bottoms     | Blue    | XS   | Spring | suppl/erd | 17.76      |
| P000016    | Shoes       | White   | M    | Spring | suppl/erd | 3157       |
| P000017    | Bottoms     | White   | S    | Spring | suppl/erb | 811        |
| P000018    | Accessories | Black   | XL   | Summer | suppl/era | 1949       |
| P000019    | Dresses     | Black   | L    | Winter | suppl/erb | 2812       |
| P000020    | Bottoms     | White   | XL   | Spring | suppl/erb | 6805       |
| P000021    | Shoes       | White   | XS   | Spring | suppl/erb | 5435       |
| P000022    | Accessories | White   | XS   | Spring | suppl/erb | 7405       |
| P000023    | Bottoms     | Red     | M    | Spring | suppl/erd | 5923       |
| P000024    | Dresses     | White   | S    | Summer | suppl/cra | 1735       |
| P000025    | Dresses     | Green   | L    | Spring | suppl/erh | 4807       |
| P000026    | Tops        | Yellow  | XS   | Fall   | suppl/erd | 4324       |
| P000027    | Tops        | Blue    | S    | Winter | suppl/era | 5373       |
| P000028    | Shoes       | Blue    | L    | Spring | suppl/erh | 1111       |
| P000029    | Bottoms     | Yellow  | M    | Fall   | suppl/erd | 6417       |
| P000030    | Bottoms     | Black   | S    | Fall   | suppl/erd | 2834       |

Kami juga melakukan Replaced Value pada kolom category yang awalnya terdapat ??? kami ubah semua menjadi Unknown, setelah itu juga tetep melakukan trimmed text agar tidak ada spasi yang berlebih.

- Working with Dates

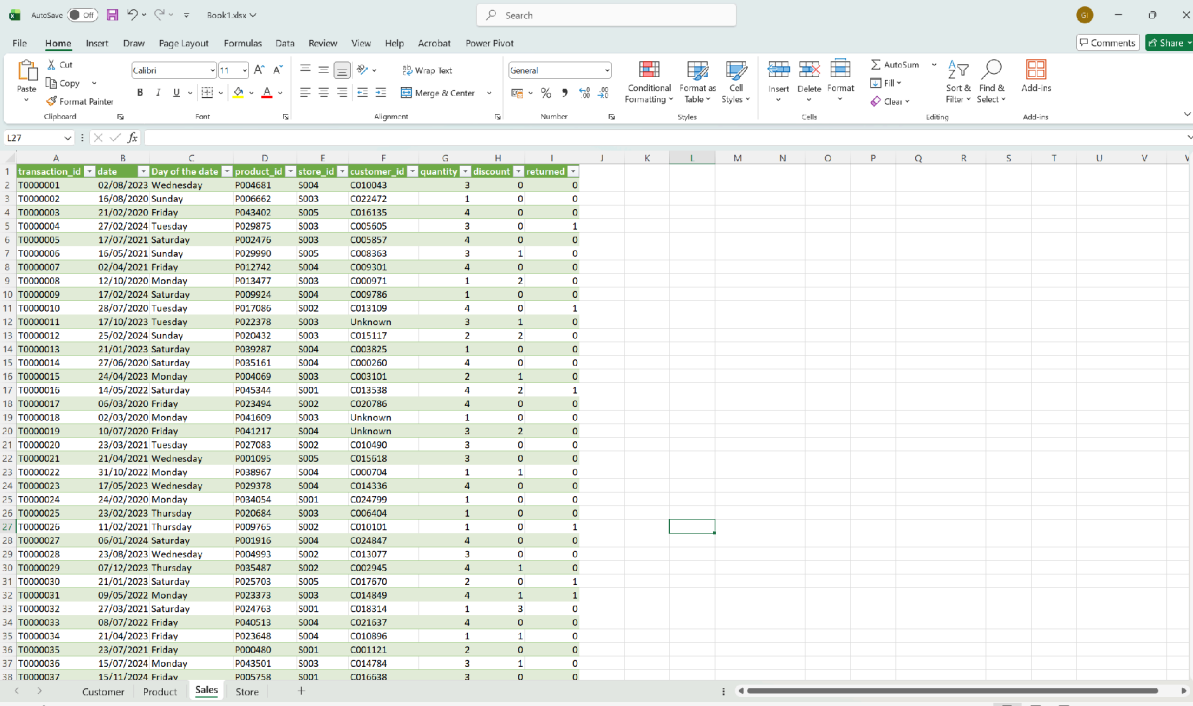
Formula Bar: `= Table.ReorderColumns(*Renamed Columns,{"transaction_id", "date", "day of the date", "product_id", "store_id", "customer_id", "quantity"})`

| transaction_id | date       | day of the date | product_id | store_id | customer_id | quantity |
|----------------|------------|-----------------|------------|----------|-------------|----------|
| T0000001       | 02/08/2020 | Wednesday       | P004981    | S004     | C002043     |          |
| T0000002       | 16/08/2020 | Sunday          | P006602    | S003     | C022472     |          |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C006135     |          |
| T0000004       | 27/02/2024 | Tuesday         | P022875    | S003     | C005905     |          |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     |          |
| T0000006       | 16/05/2021 | Sunday          | P002990    | S005     | C008363     |          |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C003061     |          |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     |          |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     |          |
| T0000010       | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     |          |
| T0000011       | 12/10/2021 | Monday          | P017138    | S003     | Unknown     |          |
| T0000012       | 28/02/2024 | Sunday          | P000482    | S003     | C005117     |          |
| T0000013       | 21/01/2023 | Saturday        | P003287    | S004     | C000325     |          |
| T0000014       | 11/06/2020 | Saturday        | P010161    | S004     | C000140     |          |
| T0000015       | 24/04/2023 | Monday          | P004609    | S003     | C001101     |          |
| T0000016       | 14/05/2022 | Saturday        | P013341    | S001     | C013338     |          |
| T0000017       | 06/11/2020 | Monday          | P011494    | S007     | C001096     |          |
| T0000018       | 02/03/2020 | Monday          | P016609    | S003     | Unknown     |          |
| T0000019       | 10/07/2020 | Friday          | P011217    | S004     | Unknown     |          |
| T0000020       | 21/01/2021 | Monday          | P010061    | S007     | C010490     |          |
| T0000021       | 21/04/2021 | Wednesday       | P010695    | S005     | C005518     |          |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     |          |
| T0000023       | 17/05/2023 | Wednesday       | P020378    | S004     | C014386     |          |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     |          |
| T0000025       | 28/02/2023 | Thursday        | P020694    | S003     | C006004     |          |
| T0000026       | 11/02/2021 | Thursday        | P007605    | S002     | C001061     |          |
| T0000027       | 06/11/2024 | Saturday        | P010161    | S004     | C010490     |          |
| T0000028       | 29/08/2023 | Wednesday       | P004993    | S002     | C003077     |          |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002245     |          |
| T0000030       | 21/01/2023 | Saturday        | P027703    | S005     | C007570     |          |
| T0000031       | 00/02/2023 | Monday          | P013372    | S003     | C010490     |          |

Pada tahap *working with dates*, dilakukan pengolahan data tanggal transaksi untuk menghasilkan informasi tambahan berupa nama hari transaksi. Proses ini dilakukan menggunakan fitur Date → Day → Name of Day pada Power Query. Hasil pengolahan tersebut menghasilkan data seperti Monday, Tuesday, Wednesday, dan seterusnya berdasarkan tanggal transaksi yang tersedia pada dataset.

Penambahan informasi hari transaksi bertujuan untuk mempermudah analisis pola penjualan berdasarkan hari tertentu. Dengan adanya atribut ini, perusahaan dapat mengetahui kapan aktivitas transaksi cenderung lebih tinggi atau lebih rendah. Informasi tersebut dapat digunakan sebagai dasar dalam pengambilan keputusan bisnis, seperti penentuan strategi promosi, pengelolaan stok produk, maupun evaluasi performa penjualan pada hari tertentu. Serta juga tetap melakukan trim agar tidak ada spasi yang berlebih.

## 2.Named Ranged and Tables



| Transaction_Id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|
| T0000001       | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        |
| T0000002       | 15/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005505     | 3        | 0        | 1        |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005957     | 4        | 0        | 0        |
| T0000006       | 16/09/2024 | Sunday          | P009990    | S005     | C003363     | 3        | 1        | 0        |
| T0000007       | 02/04/2023 | Friday          | P012742    | S004     | C003301     | 4        | 0        | 0        |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 2        | 0        |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        |
| T0000010       | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        |
| T0000012       | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 2        | 0        |
| T0000013       | 21/01/2023 | Saturday        | P039387    | S004     | C003025     | 1        | 0        | 0        |
| T0000014       | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 1        | 0        |
| T0000016       | 14/05/2022 | Saturday        | P045844    | S001     | C013338     | 4        | 2        | 1        |
| T0000017       | 06/03/2020 | Friday          | P023494    | S002     | C020786     | 4        | 0        | 0        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        |
| T0000019       | 19/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        |
| T0000021       | 21/04/2021 | Wednesday       | P001095    | S005     | C015018     | 3        | 0        | 0        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 1        | 0        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        |
| T0000027       | 06/02/2024 | Saturday        | P001916    | S004     | C024847     | 4        | 0        | 0        |
| T0000028       | 23/08/2023 | Wednesday       | P004953    | S002     | C011077     | 3        | 0        | 0        |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 1        | 0        |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        |
| T0000031       | 09/09/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        |
| T0000033       | 08/07/2022 | Friday          | P040513    | S004     | C021837     | 4        | 0        | 0        |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010896     | 1        | 1        | 0        |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        |

Semua table sudah dalam bentuk excel table.

Excel screenshot showing a data table with columns: transaction\_id, date, Day of the date, product\_id, store\_id, customer\_id, quantity, discount, returned. A 'New Name' dialog box is open, defining a name for the range G2:V2 as 'quantity'.

| transaction_id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|
| T0000001       | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        |
| T0000002       | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0        | 1        |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        |
| T0000006       | 16/05/2021 | Sunday          | P029950    | S005     | C008363     | 3        | 1        | 0        |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0        | 0        |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 2        | 0        |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        |
| T0000010       | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        |
| T0000012       | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 2        | 0        |
| T0000013       | 21/01/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0        | 0        |
| T0000014       | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 1        | 0        |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 2        | 1        |
| T0000017       | 06/03/2020 | Friday          | P023404    | S002     | C020786     | 4        | 0        | 0        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        |
| T0000021       | 21/04/2021 | Wednesday       | P001095    | S005     | C015618     | 3        | 0        | 0        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 1        | 0        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        |
| T0000027       | 06/01/2024 | Saturday        | P001916    | S004     | C024447     | 4        | 0        | 0        |
| T0000028       | 23/08/2023 | Wednesday       | P004953    | S002     | C013077     | 3        | 0        | 0        |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 1        | 0        |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        |
| T0000031       | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        |
| T0000033       | 06/07/2023 | Friday          | P040513    | S004     | C021837     | 4        | 0        | 0        |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010926     | 1        | 1        | 0        |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        |

Excel screenshot showing the same data table. A formula bar shows the formula =SUM(quantity) entered in cell Y2. A tooltip for the SUM function is visible, showing the syntax =SUM(number1, [number2], ...).

| transaction_id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|
| T0000001       | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        |
| T0000002       | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0        | 1        |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        |
| T0000006       | 16/05/2021 | Sunday          | P029950    | S005     | C008363     | 3        | 1        | 0        |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0        | 0        |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 2        | 0        |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        |
| T0000010       | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        |
| T0000012       | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 2        | 0        |
| T0000013       | 21/01/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0        | 0        |
| T0000014       | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 1        | 0        |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 2        | 1        |
| T0000017       | 06/03/2020 | Friday          | P023404    | S002     | C020786     | 4        | 0        | 0        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        |
| T0000021       | 21/04/2021 | Wednesday       | P001095    | S005     | C015618     | 3        | 0        | 0        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 1        | 0        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        |
| T0000027       | 06/01/2024 | Saturday        | P001916    | S004     | C024447     | 4        | 0        | 0        |
| T0000028       | 23/08/2023 | Wednesday       | P004953    | S002     | C013077     | 3        | 0        | 0        |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 1        | 0        |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        |
| T0000031       | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        |
| T0000033       | 06/07/2023 | Friday          | P040513    | S004     | C021837     | 4        | 0        | 0        |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010926     | 1        | 1        | 0        |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        |

|                                                                                               |                     |          |   |
|-----------------------------------------------------------------------------------------------|---------------------|----------|---|
| <div> <span>✕</span> <span>✓</span> <span><i>fx</i></span> <span>=SUM(quantity)</span> </div> |                     |          |   |
|                                                                                               | C                   | D        | E |
|                                                                                               | Price Change (%)    | 0%       |   |
|                                                                                               | Total Quantity Sold | 124076   |   |
|                                                                                               | Skenario            | 124076   |   |
|                                                                                               | -15%                | 105464,6 |   |
|                                                                                               | -10%                | 111668,4 |   |
|                                                                                               | -5%                 | 117872,2 |   |
|                                                                                               | 0%                  | 124076   |   |
|                                                                                               | 5%                  | 130279,8 |   |
|                                                                                               | 10%                 | 136483,6 |   |
|                                                                                               | 15%                 | 142687,4 |   |

## 1. Quantity

Named Range QuantityData dibuat berdasarkan kolom quantity pada tabel penjualan (sales\_messy). Kolom ini berisi jumlah produk yang dibeli dalam setiap transaksi.

Named Range ini digunakan untuk menghitung total produk yang terjual, melakukan analisis jumlah pembelian pelanggan, serta mendukung pembuatan visualisasi dashboard terkait tren penjualan produk.

Contoh penggunaan formula:

=SUM(Quantity)

Formula tersebut digunakan untuk menghitung total keseluruhan produk yang berhasil terjual.

| Transaction_id | Date       | Day of the date | Product_id | Store_id | Customer_id | Quantity | Discount | Returned |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|
| T0000001       | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        |
| T0000002       | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005505     | 3        | 0        | 1        |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        |
| T0000006       | 16/09/2021 | Sunday          | P029990    | S005     | C006368     | 3        | 1        | 0        |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0        | 0        |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 2        | 0        |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        |
| T0000010       | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        |
| T0000012       | 25/02/2024 | Sunday          | P000432    | S003     | C003117     | 2        | 2        | 0        |
| T0000013       | 21/01/2023 | Saturday        | P030267    | S004     | C003255     | 1        | 0        | 0        |
| T0000014       | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 1        | 0        |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 2        | 1        |
| T0000017       | 06/03/2020 | Friday          | P023494    | S002     | C020786     | 4        | 0        | 0        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        |
| T0000021       | 21/04/2021 | Wednesday       | P001055    | S005     | C015518     | 3        | 0        | 0        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 1        | 0        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        |
| T0000027       | 06/02/2024 | Saturday        | P001915    | S004     | C024947     | 4        | 0        | 0        |
| T0000028       | 23/08/2023 | Wednesday       | P004999    | S002     | C013077     | 3        | 0        | 0        |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 1        | 0        |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        |
| T0000031       | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        |
| T0000033       | 08/07/2022 | Friday          | P040513    | S004     | C021837     | 4        | 0        | 0        |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010996     | 1        | 1        | 0        |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        |

## 2. Returned

Named Range ReturnedData dibuat berdasarkan kolom returned pada tabel penjualan (sales\_messy). Kolom ini menggunakan tipe data biner, yaitu:

- 0 = produk tidak dikembalikan
- 1 = produk dikembalikan

Named Range ini digunakan untuk menganalisis jumlah transaksi pengembalian barang (return) dan membantu perusahaan memahami tingkat retur produk.

Contoh penggunaan formula:

=SUM(Returned)

Formula tersebut digunakan untuk menghitung total transaksi produk yang dikembalikan oleh pelanggan.

### 3.Logical, Lookup, Conditional Functions

The screenshot displays the Microsoft Excel interface with a data table. The formula bar at the top shows the logical function: `=IF([@returned];"Returned";"Not Returned")`. The table contains 38 rows of transaction data, including transaction ID, date, day of the week, product ID, store ID, customer ID, quantity, discount, returned status, returned info, and category.

| transaction_id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned | returned_info | category    |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|---------------|-------------|
| T0000001       | 02/08/2023 | Wednesday       | P004661    | S004     | C010043     | 3        | 0        | 0        | Not Returned  | Shoes       |
| T0000002       | 16/08/2020 | Sunday          | P006663    | S003     | C027472     | 1        | 0        | 0        | Not Returned  | Accessories |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        | Not Returned  | Shoes       |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0        | 1        | Returned      | Shoes       |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000006       | 16/05/2021 | Sunday          | P029900    | S005     | C008363     | 3        | 1        | 0        | Not Returned  | Accessories |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0        | 0        | Not Returned  | Bottoms     |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C009071     | 1        | 2        | 0        | Not Returned  | Accessories |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000010       | 28/07/2020 | Tuesday         | P017066    | S002     | C013109     | 4        | 0        | 1        | Returned      | Dresses     |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        | Not Returned  | Accessories |
| T0000012       | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 2        | 0        | Not Returned  | Shoes       |
| T0000013       | 21/01/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000014       | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000015       | 12/10/2020 | Monday          | P004068    | S003     | C003101     | 1        | 0        | 0        | Not Returned  | Accessories |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 2        | 1        | Returned      | Tops        |
| T0000017       | 06/03/2020 | Friday          | P023404    | S002     | C020786     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        | Not Returned  | Shoes       |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        | Not Returned  | Bottoms     |
| T0000021       | 21/04/2021 | Wednesday       | P001095    | S005     | C015528     | 3        | 0        | 0        | Not Returned  | Tops        |
| T0000022       | 31/10/2022 | Monday          | P039867    | S004     | C000704     | 1        | 1        | 0        | Not Returned  | Tops        |
| T0000023       | 17/06/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        | Not Returned  | Dresses     |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        | Not Returned  | Bottoms     |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        | Returned      | Dresses     |
| T0000027       | 06/01/2024 | Saturday        | P001916    | S004     | C024847     | 4        | 0        | 0        | Not Returned  | Accessories |
| T0000028       | 23/09/2023 | Wednesday       | P004093    | S002     | C011077     | 3        | 0        | 0        | Not Returned  | Accessories |
| T0000029       | 07/13/2023 | Thursday        | P035487    | S002     | C003045     | 4        | 1        | 0        | Not Returned  | Shoes       |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        | Returned      | Tops        |
| T0000031       | 09/09/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        | Returned      | Tops        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        | Not Returned  | Shoes       |
| T0000033       | 08/07/2022 | Friday          | P040513    | S004     | C021637     | 4        | 0        | 0        | Not Returned  | Bottoms     |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010936     | 1        | 1        | 0        | Not Returned  | Bottoms     |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        | Not Returned  | Accessories |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        | Not Returned  | Shoes       |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        | Not Returned  | Accessories |

#### Logical Function (IF)

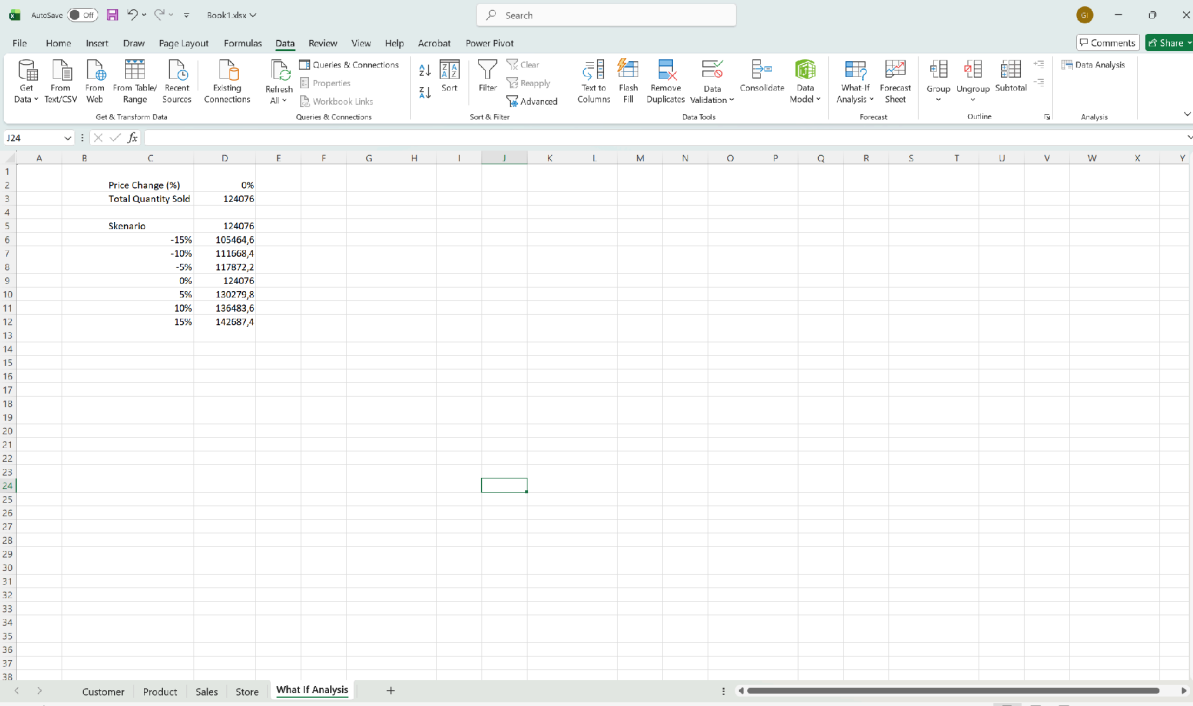
Pada proyek ini digunakan fungsi logika IF untuk mengubah data numerik pada kolom *returned* menjadi informasi yang lebih mudah dipahami. Kami melakukan logical function pada kolom baru bernama *returned\_info*. Formula tersebut digunakan untuk mengecek nilai pada kolom *returned*. Jika nilai sama dengan 1, maka Excel akan menampilkan teks “Returned”. Jika nilainya 0, maka akan ditampilkan “Not Returned”. Fungsi ini membantu proses analisis data menjadi lebih mudah dibaca dibandingkan hanya menggunakan angka biner. Hasil dari fungsi ini kemudian digunakan dalam proses analisis transaksi dan visualisasi dashboard terkait pengembalian produk.

| transaction_id | date       | day_of_the_date | product_id | store_id | customer_id | quantity | discount | returned | returned_info | category    |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|---------------|-------------|
| T0000001       | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        | Not Returned  | Shoes       |
| T0000002       | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        | Not Returned  | Accessories |
| T0000003       | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        | Not Returned  | Shoes       |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0        | 1        | Returned      | Shoes       |
| T0000005       | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000006       | 16/05/2021 | Sunday          | P029960    | S005     | C006353     | 3        | 1        | 0        | Not Returned  | Accessories |
| T0000007       | 03/04/2021 | Friday          | P013742    | S004     | C009301     | 4        | 0        | 0        | Not Returned  | Bottoms     |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 2        | 0        | Not Returned  | Accessories |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000010       | 26/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        | Returned      | Dresses     |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 1        | 0        | Not Returned  | Accessories |
| T0000012       | 25/02/2024 | Sunday          | P010432    | S003     | C015117     | 2        | 2        | 0        | Not Returned  | Shoes       |
| T0000013       | 24/04/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000014       | 27/04/2020 | Saturday        | P035151    | S004     | C000250     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 1        | 0        | Not Returned  | Accessories |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 2        | 1        | Returned      | Tops        |
| T0000017       | 06/03/2020 | Friday          | P023404    | S002     | C020786     | 4        | 0        | 0        | Not Returned  | Tops        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 2        | 0        | Not Returned  | Shoes       |
| T0000020       | 23/04/2021 | Tuesday         | P027083    | S002     | C004990     | 3        | 0        | 0        | Not Returned  | Bottoms     |
| T0000021       | 21/04/2021 | Wednesday       | P001095    | S005     | C015618     | 3        | 0        | 0        | Not Returned  | Tops        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 1        | 0        | Not Returned  | Tops        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        | Not Returned  | Dresses     |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        | Not Returned  | Tops        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        | Not Returned  | Bottoms     |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        | Returned      | Dresses     |
| T0000027       | 06/04/2024 | Saturday        | P001916    | S004     | C024647     | 4        | 0        | 0        | Not Returned  | Accessories |
| T0000028       | 23/08/2023 | Wednesday       | P004963    | S002     | C013077     | 3        | 0        | 0        | Not Returned  | Accessories |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 1        | 0        | Not Returned  | Shoes       |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        | Returned      | Tops        |
| T0000031       | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 1        | 1        | Returned      | Tops        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 3        | 0        | Not Returned  | Shoes       |
| T0000033       | 06/07/2022 | Friday          | P040513    | S004     | C021837     | 4        | 0        | 0        | Not Returned  | Bottoms     |
| T0000034       | 21/04/2023 | Friday          | P013648    | S004     | C010596     | 1        | 1        | 0        | Not Returned  | Bottoms     |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        | Not Returned  | Accessories |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 1        | 0        | Not Returned  | Shoes       |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        | Not Returned  | Accessories |

## Lookup Function (VLOOKUP)

Pada proyek ini digunakan fungsi VLOOKUP untuk mengambil informasi kategori produk dari tabel produk ke tabel penjualan berdasarkan product\_id. Formula tersebut digunakan untuk mencari nilai *product\_id* pada tabel produk, kemudian mengambil data kategori produk yang sesuai seperti Accessories, Shoes, Tops, Bottoms, dan Dresses. Fungsi VLOOKUP membantu proses penggabungan data antar tabel sehingga informasi kategori produk dapat digunakan dalam analisis penjualan dan pembuatan dashboard.

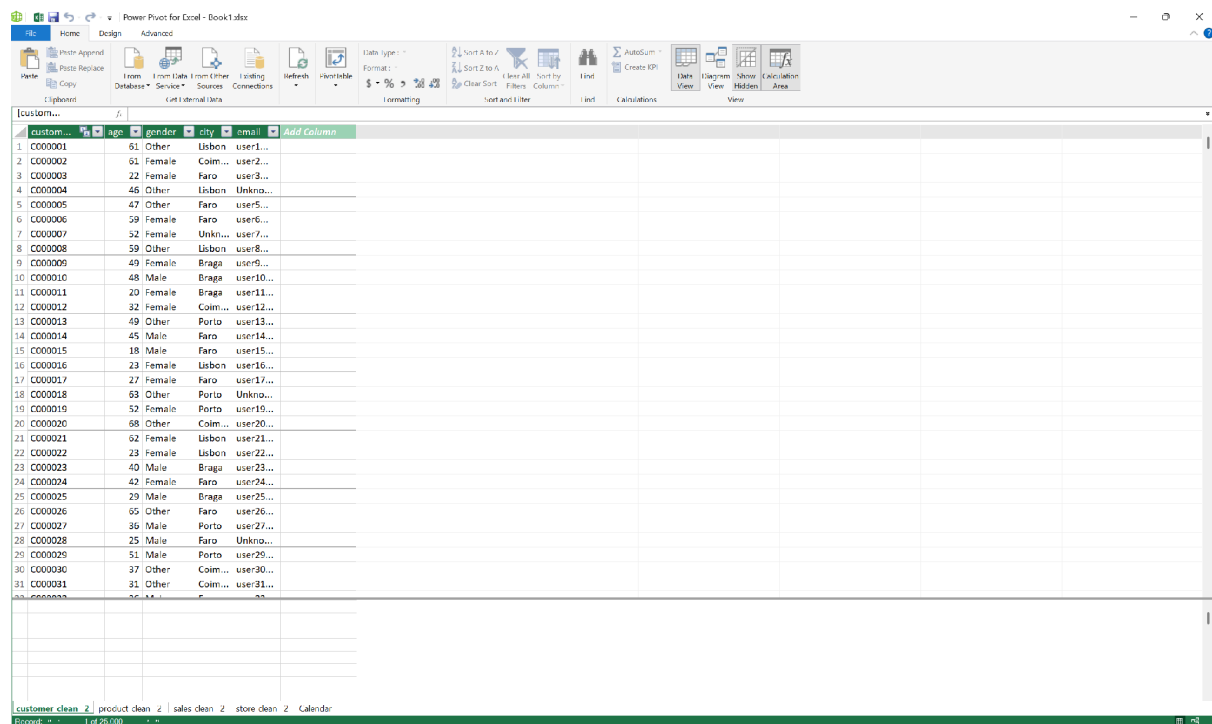
## 4. Analysis with Data Tables



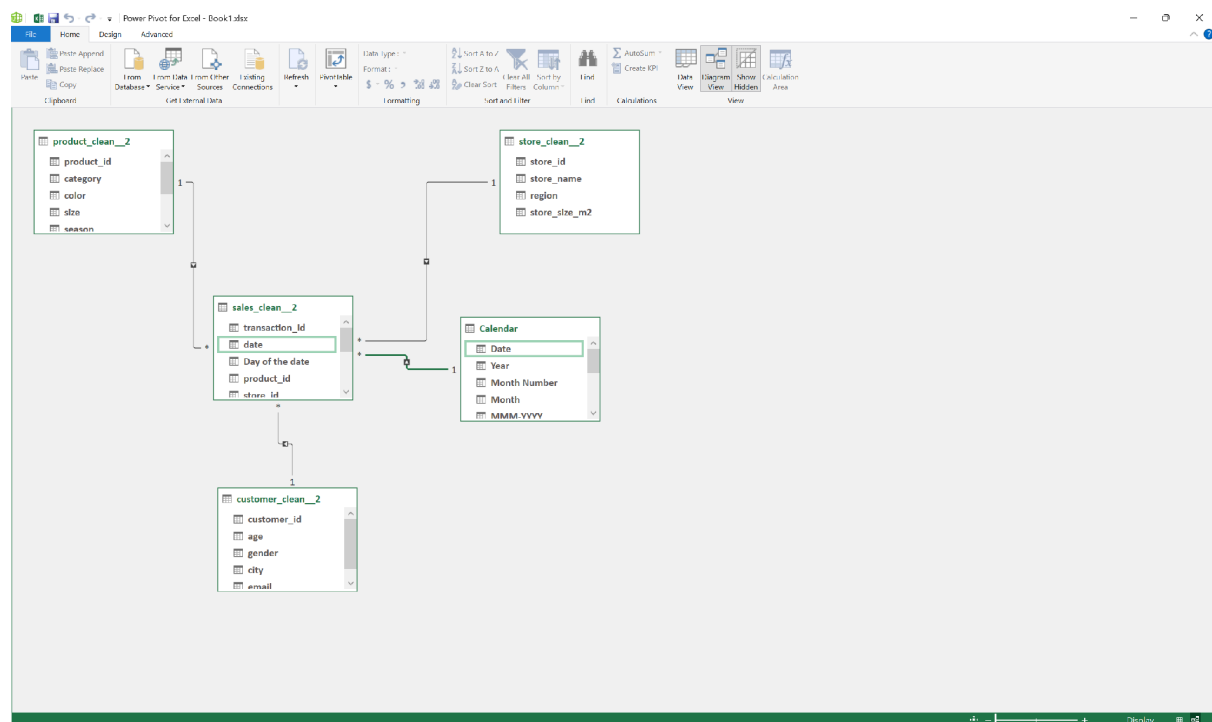
| Price Change (%) | Total Quantity Sold |
|------------------|---------------------|
| 0%               | 124076              |
| -15%             | 105464,6            |
| -10%             | 111568,4            |
| -5%              | 117872,2            |
| 0%               | 124076              |
| 5%               | 130279,8            |
| 10%              | 136483,6            |
| 15%              | 142687,4            |

What-If Analysis dilakukan untuk menganalisis dampak perubahan persentase penjualan terhadap total quantity sold pada dataset. Analisis ini menggunakan fitur Data Table pada Microsoft Excel dengan beberapa skenario perubahan, yaitu -15%, -10%, -5%, 0%, 5%, 10%, dan 15%. Nilai awal total quantity sold berdasarkan dataset adalah sebesar 124.076. Selanjutnya dilakukan simulasi menggunakan formula perhitungan projected quantity sold untuk melihat perubahan hasil penjualan pada setiap skenario. Hasil analisis menunjukkan bahwa ketika terjadi penurunan sebesar 15%, total quantity sold ikut menurun menjadi sekitar 105.464. Sebaliknya, ketika terjadi peningkatan sebesar 15%, projected quantity sold meningkat menjadi sekitar 142.687. Dari hasil tersebut dapat disimpulkan bahwa perubahan persentase penjualan memiliki pengaruh langsung terhadap jumlah produk yang terjual. Semakin besar peningkatan persentase penjualan, maka quantity sold juga meningkat, sedangkan penurunan persentase menyebabkan jumlah penjualan menurun. Analisis ini berguna untuk membantu proses perencanaan dan pengambilan keputusan bisnis berdasarkan kemungkinan perubahan kondisi penjualan di masa mendatang.

## 5. Power Pivot and The Data Model



| custom  | age | gender | city     | email     |
|---------|-----|--------|----------|-----------|
| C000001 | 61  | Other  | Lisbon   | user1...  |
| C000002 | 61  | Female | Coim...  | user2...  |
| C000003 | 22  | Female | Faro     | user3...  |
| C000004 | 46  | Other  | Lisbon   | Unkno...  |
| C000005 | 47  | Other  | Faro     | user5...  |
| C000006 | 59  | Female | Faro     | user6...  |
| C000007 | 32  | Female | Unkno... | user7...  |
| C000008 | 59  | Other  | Lisbon   | user8...  |
| C000009 | 49  | Female | Braga    | user9...  |
| C000010 | 48  | Male   | Braga    | user10... |
| C000011 | 29  | Female | Braga    | user11... |
| C000012 | 32  | Female | Coim...  | user12... |
| C000013 | 49  | Other  | Porto    | user13... |
| C000014 | 45  | Male   | Faro     | user14... |
| C000015 | 18  | Male   | Faro     | user15... |
| C000016 | 23  | Female | Lisbon   | user16... |
| C000017 | 27  | Female | Faro     | user17... |
| C000018 | 63  | Other  | Porto    | Unkno...  |
| C000019 | 52  | Female | Porto    | user18... |
| C000020 | 68  | Other  | Coim...  | user20... |
| C000021 | 62  | Female | Lisbon   | user21... |
| C000022 | 23  | Female | Lisbon   | user22... |
| C000023 | 40  | Male   | Braga    | user23... |
| C000024 | 42  | Female | Faro     | user24... |
| C000025 | 29  | Male   | Braga    | user25... |
| C000026 | 66  | Other  | Faro     | user26... |
| C000027 | 36  | Male   | Porto    | user27... |
| C000028 | 25  | Male   | Faro     | Unkno...  |
| C000029 | 51  | Male   | Porto    | user28... |
| C000030 | 37  | Other  | Coim...  | user30... |
| C000031 | 31  | Other  | Coim...  | user31... |



Pada tahap ini dilakukan proses pengolahan data menggunakan fitur Power Pivot pada Microsoft Excel. Langkah pertama yang dilakukan adalah mengaktifkan Power Pivot agar dataset dapat dimasukkan ke dalam Data Model. Setelah data dimasukkan ke dalam Data Model, dilakukan pembuatan relationship antar tabel menggunakan kolom yang memiliki keterkaitan.

Power Pivot for Excel - Book1.xlsx

Home

Design

Advanced

Insert

Table

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`=CUBEVALUE("ThisWorkbookDataModel";"[Measures].[Total Returned]")`

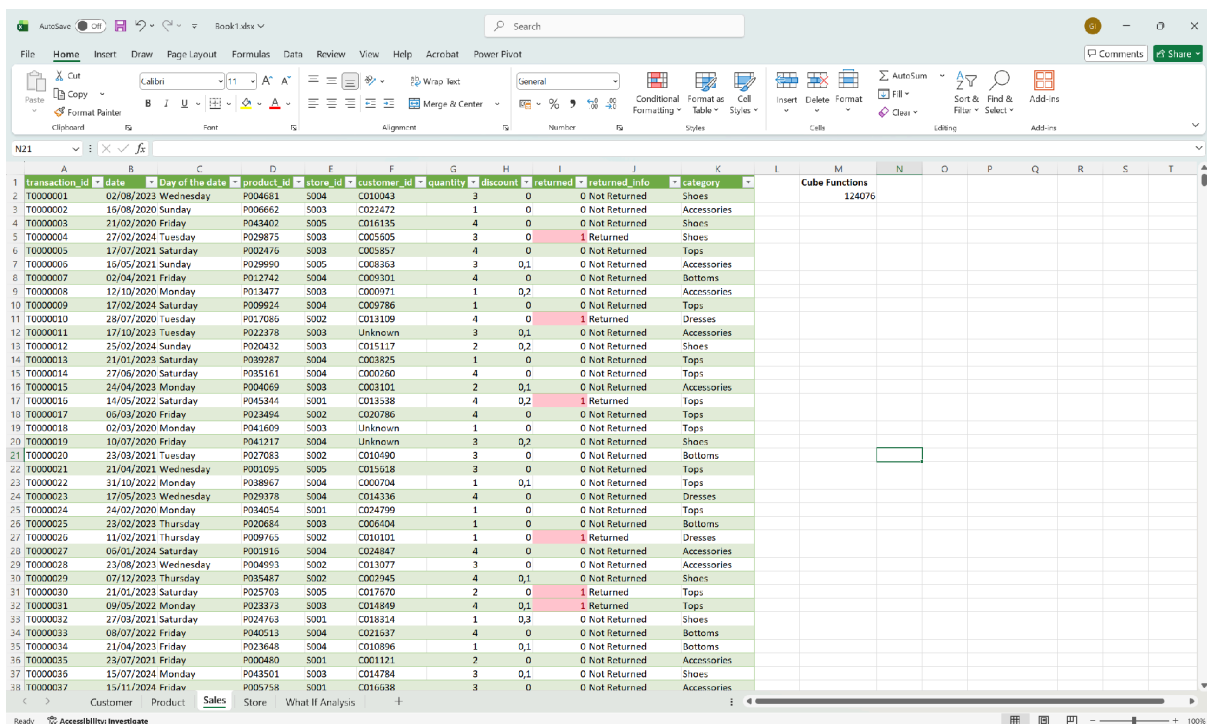
|   | A | B | C | D           | E | F | G | H | I | J                   | K | L | M | N | O | P | Q              | R | S | T | U | V | W | X |
|---|---|---|---|-------------|---|---|---|---|---|---------------------|---|---|---|---|---|---|----------------|---|---|---|---|---|---|---|
| 1 |   |   |   | Average Age |   |   |   |   |   | Total Quantity Sold |   |   |   |   |   |   | Total Returned |   |   |   |   |   |   |   |
| 2 |   |   |   | 42,46128    |   |   |   |   |   | 124076              |   |   |   |   |   |   | 4919           |   |   |   |   |   |   |   |
| 3 |   |   |   |             |   |   |   |   |   |                     |   |   |   |   |   |   |                |   |   |   |   |   |   |   |

Selain membuat relationship, dilakukan juga pembuatan calculated field atau measure menggunakan bahasa DAX (Data Analysis Expressions). Measure pertama adalah Total Quantity Sold yang digunakan untuk menghitung total jumlah produk yang terjual menggunakan fungsi SUM pada kolom quantity. Measure kedua adalah Total Returned yang digunakan untuk menghitung jumlah produk yang mengalami return berdasarkan kolom returned yang berisi nilai 0 dan 1. Measure ini membantu dalam menganalisis performa penjualan dan tingkat pengembalian produk.

Selanjutnya digunakan Cube Function untuk mengambil hasil perhitungan dari Power Pivot Data Model ke worksheet Excel biasa. Fungsi yang digunakan adalah CUBEVALUE, yang berfungsi menampilkan hasil measure secara langsung ke worksheet sehingga mempermudah proses analisis dan visualisasi data.

## 6. Conditional and Number Formatting

- Conditional Formatting



| transaction_id | date       | day_of_the_week | product_id | store_id | customer_id | quantity | discount | returned | returned_info  | category    |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|----------------|-------------|
| T0000001       | 02/08/2023 | Wednesday       | P004651    | S004     | C010043     | 3        | 0        | 0        | 0 Not Returned | Shoes       |
| T0000002       | 16/08/2020 | Sunday          | P006662    | S003     | C023472     | 1        | 0        | 0        | 0 Not Returned | Accessories |
| T0000003       | 21/03/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        | 0 Not Returned | Shoes       |
| T0000004       | 27/02/2024 | Tuesday         | P029875    | S003     | C005505     | 3        | 0        | 1        | 1 Returned     | Shoes       |
| T0000005       | 17/01/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| T0000006       | 16/05/2021 | Sunday          | P029990    | S005     | C008363     | 3        | 0,1      | 0        | 0 Not Returned | Accessories |
| T0000007       | 02/04/2021 | Friday          | P012742    | S004     | C000301     | 4        | 0        | 0        | 0 Not Returned | Bottoms     |
| T0000008       | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 0,2      | 0        | 0 Not Returned | Accessories |
| T0000009       | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| T0000010       | 28/01/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        | 1 Returned     | Dresses     |
| T0000011       | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 0,1      | 0        | 0 Not Returned | Accessories |
| T0000012       | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 0,2      | 0        | 0 Not Returned | Shoes       |
| T0000013       | 21/01/2023 | Saturday        | P0309287   | S004     | C003825     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| T0000014       | 27/04/2020 | Saturday        | P035161    | S004     | C000760     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| T0000015       | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 0,1      | 0        | 0 Not Returned | Accessories |
| T0000016       | 14/05/2022 | Saturday        | P045344    | S001     | C013338     | 4        | 0,2      | 1        | 1 Returned     | Tops        |
| T0000017       | 06/03/2022 | Friday          | P023494    | S002     | C020786     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| T0000018       | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| T0000019       | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 0,2      | 0        | 0 Not Returned | Shoes       |
| T0000020       | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        | 0 Not Returned | Bottoms     |
| T0000021       | 23/04/2021 | Wednesday       | P031095    | S005     | C013538     | 3        | 0        | 0        | 0 Not Returned | Tops        |
| T0000022       | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 0,1      | 0        | 0 Not Returned | Tops        |
| T0000023       | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        | 0 Not Returned | Dresses     |
| T0000024       | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| T0000025       | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        | 0 Not Returned | Bottoms     |
| T0000026       | 11/02/2021 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        | 1 Returned     | Dresses     |
| T0000027       | 06/02/2024 | Saturday        | P001916    | S004     | C024847     | 4        | 0        | 0        | 0 Not Returned | Accessories |
| T0000028       | 23/08/2023 | Wednesday       | P004953    | S002     | C013077     | 3        | 0        | 0        | 0 Not Returned | Accessories |
| T0000029       | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 0,1      | 0        | 0 Not Returned | Shoes       |
| T0000030       | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        | 1 Returned     | Tops        |
| T0000031       | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 0,1      | 1        | 1 Returned     | Tops        |
| T0000032       | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 0,3      | 0        | 0 Not Returned | Shoes       |
| T0000033       | 08/07/2022 | Friday          | P040513    | S004     | C021937     | 4        | 0        | 0        | 0 Not Returned | Bottoms     |
| T0000034       | 21/04/2023 | Friday          | P023648    | S004     | C010896     | 1        | 0,1      | 0        | 0 Not Returned | Bottoms     |
| T0000035       | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        | 0 Not Returned | Accessories |
| T0000036       | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 0,1      | 0        | 0 Not Returned | Shoes       |
| T0000037       | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        | 0 Not Returned | Accessories |

Conditional Formatting diterapkan pada kolom returned untuk menyoroti transaksi yang memiliki nilai 1, yang menandakan produk dikembalikan (returned). Aturan formatting dibuat agar sel dengan nilai 1 otomatis diberi warna kuning sehingga transaksi return dapat lebih mudah diidentifikasi saat proses analisis data dilakukan. Penerapan formatting ini membantu meningkatkan keterbacaan data dan mempermudah proses monitoring transaksi pengembalian barang.

- Number Formatting

| transaction_id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned | returned_info  | category    |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|----------------|-------------|
| 1              | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0        | 0        | 0 Not Returned | Shoes       |
| 2              | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0        | 0        | 0 Not Returned | Accessories |
| 3              | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0        | 0        | 0 Not Returned | Shoes       |
| 4              | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0        | 1        | 1 Returned     | Shoes       |
| 5              | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| 6              | 16/05/2021 | Sunday          | P029990    | S005     | C008363     | 3        | 0.1      | 0        | 0 Not Returned | Accessories |
| 7              | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0        | 0        | 0 Not Returned | Bottoms     |
| 8              | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 0.2      | 0        | 0 Not Returned | Accessories |
| 9              | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| 10             | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0        | 1        | 1 Returned     | Dresses     |
| 11             | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 0.1      | 0        | 0 Not Returned | Accessories |
| 12             | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 0.2      | 0        | 0 Not Returned | Shoes       |
| 13             | 21/01/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| 14             | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| 15             | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 0.1      | 0        | 0 Not Returned | Accessories |
| 16             | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 0.2      | 1        | 1 Returned     | Tops        |
| 17             | 06/03/2020 | Friday          | P023484    | S002     | C020786     | 4        | 0        | 0        | 0 Not Returned | Tops        |
| 18             | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| 19             | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 0.2      | 0        | 0 Not Returned | Shoes       |
| 20             | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0        | 0        | 0 Not Returned | Bottoms     |
| 21             | 20/00/2021 | Wednesday       | P001095    | S005     | C015618     | 3        | 0        | 0        | 0 Not Returned | Tops        |
| 22             | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 0.1      | 0        | 0 Not Returned | Tops        |
| 23             | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0        | 0        | 0 Not Returned | Dresses     |
| 24             | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0        | 0        | 0 Not Returned | Tops        |
| 25             | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0        | 0        | 0 Not Returned | Bottoms     |
| 26             | 11/02/2022 | Thursday        | P009765    | S002     | C010101     | 1        | 0        | 1        | 1 Returned     | Dresses     |
| 27             | 06/01/2024 | Saturday        | P001916    | S004     | C024847     | 4        | 0        | 0        | 0 Not Returned | Accessories |
| 28             | 23/08/2023 | Wednesday       | P004993    | S002     | C013077     | 3        | 0        | 0        | 0 Not Returned | Accessories |
| 29             | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 0.1      | 0        | 0 Not Returned | Shoes       |
| 30             | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0        | 1        | 1 Returned     | Tops        |
| 31             | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 0.1      | 1        | 1 Returned     | Tops        |
| 32             | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 0.3      | 0        | 0 Not Returned | Shoes       |
| 33             | 08/07/2022 | Friday          | P040513    | S004     | C021637     | 4        | 0        | 0        | 0 Not Returned | Bottoms     |
| 34             | 21/04/2023 | Friday          | P023648    | S004     | C010896     | 1        | 0.1      | 0        | 0 Not Returned | Bottoms     |
| 35             | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0        | 0        | 0 Not Returned | Accessories |
| 36             | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 0.1      | 0        | 0 Not Returned | Shoes       |
| 37             | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0        | 0        | 0 Not Returned | Accessories |

| transaction_id | date       | Day of the date | product_id | store_id | customer_id | quantity | discount | returned | returned_info  | category    |
|----------------|------------|-----------------|------------|----------|-------------|----------|----------|----------|----------------|-------------|
| 1              | 02/08/2023 | Wednesday       | P004681    | S004     | C010043     | 3        | 0%       | 0        | 0 Not Returned | Shoes       |
| 2              | 16/08/2020 | Sunday          | P006662    | S003     | C022472     | 1        | 0%       | 0        | 0 Not Returned | Accessories |
| 3              | 21/02/2020 | Friday          | P043402    | S005     | C016135     | 4        | 0%       | 0        | 0 Not Returned | Shoes       |
| 4              | 27/02/2024 | Tuesday         | P029875    | S003     | C005605     | 3        | 0%       | 1        | 1 Returned     | Shoes       |
| 5              | 17/07/2021 | Saturday        | P002476    | S003     | C005857     | 4        | 0%       | 0        | 0 Not Returned | Tops        |
| 6              | 16/05/2021 | Sunday          | P029990    | S005     | C008363     | 3        | 10%      | 0        | 0 Not Returned | Accessories |
| 7              | 02/04/2021 | Friday          | P012742    | S004     | C009301     | 4        | 0%       | 0        | 0 Not Returned | Bottoms     |
| 8              | 12/10/2020 | Monday          | P013477    | S003     | C000971     | 1        | 20%      | 0        | 0 Not Returned | Accessories |
| 9              | 17/02/2024 | Saturday        | P009924    | S004     | C009786     | 1        | 0%       | 0        | 0 Not Returned | Tops        |
| 10             | 28/07/2020 | Tuesday         | P017086    | S002     | C013109     | 4        | 0%       | 1        | 1 Returned     | Dresses     |
| 11             | 17/10/2023 | Tuesday         | P022378    | S003     | Unknown     | 3        | 10%      | 0        | 0 Not Returned | Accessories |
| 12             | 25/02/2024 | Sunday          | P020432    | S003     | C015117     | 2        | 20%      | 0        | 0 Not Returned | Shoes       |
| 13             | 21/01/2023 | Saturday        | P039287    | S004     | C003825     | 1        | 0%       | 0        | 0 Not Returned | Tops        |
| 14             | 27/06/2020 | Saturday        | P035161    | S004     | C000260     | 4        | 0%       | 0        | 0 Not Returned | Tops        |
| 15             | 24/04/2023 | Monday          | P004069    | S003     | C003101     | 2        | 10%      | 0        | 0 Not Returned | Accessories |
| 16             | 14/05/2022 | Saturday        | P045344    | S001     | C013538     | 4        | 20%      | 1        | 1 Returned     | Tops        |
| 17             | 06/03/2020 | Friday          | P023484    | S002     | C020786     | 4        | 0%       | 0        | 0 Not Returned | Tops        |
| 18             | 02/03/2020 | Monday          | P041609    | S003     | Unknown     | 1        | 0%       | 0        | 0 Not Returned | Tops        |
| 19             | 10/07/2020 | Friday          | P041217    | S004     | Unknown     | 3        | 20%      | 0        | 0 Not Returned | Shoes       |
| 20             | 23/03/2021 | Tuesday         | P027083    | S002     | C010490     | 3        | 0%       | 0        | 0 Not Returned | Bottoms     |
| 21             | 20/00/2021 | Wednesday       | P001095    | S005     | C015618     | 3        | 0%       | 0        | 0 Not Returned | Tops        |
| 22             | 31/10/2022 | Monday          | P038967    | S004     | C000704     | 1        | 10%      | 0        | 0 Not Returned | Tops        |
| 23             | 17/05/2023 | Wednesday       | P029378    | S004     | C014336     | 4        | 0%       | 0        | 0 Not Returned | Dresses     |
| 24             | 24/02/2020 | Monday          | P034054    | S001     | C024799     | 1        | 0%       | 0        | 0 Not Returned | Tops        |
| 25             | 23/02/2023 | Thursday        | P020684    | S003     | C006404     | 1        | 0%       | 0        | 0 Not Returned | Bottoms     |
| 26             | 11/02/2022 | Thursday        | P009765    | S002     | C010101     | 1        | 0%       | 1        | 1 Returned     | Dresses     |
| 27             | 06/01/2024 | Saturday        | P001916    | S004     | C024847     | 4        | 0%       | 0        | 0 Not Returned | Accessories |
| 28             | 23/08/2023 | Wednesday       | P004993    | S002     | C013077     | 3        | 0%       | 0        | 0 Not Returned | Accessories |
| 29             | 07/12/2023 | Thursday        | P035487    | S002     | C002945     | 4        | 10%      | 0        | 0 Not Returned | Shoes       |
| 30             | 21/01/2023 | Saturday        | P025703    | S005     | C017670     | 2        | 0%       | 1        | 1 Returned     | Tops        |
| 31             | 09/05/2022 | Monday          | P023373    | S003     | C014849     | 4        | 10%      | 1        | 1 Returned     | Tops        |
| 32             | 27/03/2021 | Saturday        | P024763    | S001     | C018314     | 1        | 30%      | 0        | 0 Not Returned | Shoes       |
| 33             | 08/07/2022 | Friday          | P040513    | S004     | C021637     | 4        | 0%       | 0        | 0 Not Returned | Bottoms     |
| 34             | 21/04/2023 | Friday          | P023648    | S004     | C010896     | 1        | 10%      | 0        | 0 Not Returned | Bottoms     |
| 35             | 23/07/2021 | Friday          | P000480    | S001     | C001121     | 2        | 0%       | 0        | 0 Not Returned | Accessories |
| 36             | 15/07/2024 | Monday          | P043501    | S003     | C014784     | 3        | 10%      | 0        | 0 Not Returned | Shoes       |
| 37             | 15/11/2024 | Friday          | P005758    | S001     | C016638     | 3        | 0%       | 0        | 0 Not Returned | Accessories |

Number Formatting diterapkan pada kolom discount dengan mengubah tampilan angka menjadi format persentase (%). Formatting ini dilakukan agar nilai diskon lebih mudah dibaca dan dipahami dalam bentuk persentase, seperti 10%, 20%, dan 30%, dibandingkan angka desimal biasa. Dengan penggunaan format persentase, proses analisis data penjualan menjadi lebih jelas, informatif, dan mudah diinterpretasikan.

- Calculated Fields and Measures

The screenshot shows a Microsoft Excel window titled 'Power Pivot for Excel - Book1.xlsx'. The ribbon is set to 'PivotTable Tools' with the 'Formulas' tab selected. A PivotTable is displayed with the following data:

| customer_id | age | gender | city    | email     |
|-------------|-----|--------|---------|-----------|
| 22          | 23  | Female | Lisbon  | user22... |
| 23          | 40  | Male   | Braga   | user23... |
| 24          | 42  | Female | Faro    | user24... |
| 25          | 29  | Male   | Braga   | user25... |
| 26          | 65  | Other  | Faro    | user26... |
| 27          | 36  | Male   | Porto   | user27... |
| 28          | 25  | Male   | Faro    | Unkno...  |
| 29          | 51  | Male   | Porto   | user29... |
| 30          | 37  | Other  | Coim... | user30... |
| 31          | 31  | Other  | Coim... | user31... |
| 32          | 36  | Male   | Faro    | user32... |
| 33          | 39  | Male   | Porto   | user33... |
| 34          | 26  | Male   | Lisbon  | user34... |
| 35          | 27  | Male   | Porto   | user35... |
| 36          | 54  | Male   | Porto   | user36... |
| 37          | 50  | Male   | Lisbon  | user37... |
| 38          | 40  | Other  | Lisbon  | Unkno...  |
| 39          | 67  | Male   | Lisbon  | user39... |
| 40          | 54  | Male   | Lisbon  | user40... |
| 41          | 21  | Other  | Faro    | user41... |
| 42          | 52  | Other  | Coim... | user42... |
| 43          | 55  | Female | Lisbon  | user43... |
| 44          | 35  | Male   | Faro    | user44... |
| 45          | 51  | Other  | Coim... | user45... |
| 46          | 64  | Male   | Porto   | user46... |
| 47          | 22  | Male   | Coim... | user47... |
| 48          | 19  | Male   | Porto   | user48... |
| 49          | 38  | Female | Braga   | user49... |
| 50          | 29  | Other  | Porto   | user50... |
| 51          | 44  | Other  | Braga   | user51... |
| 52          | 39  | Female | Porto   | user52... |

Below the data, a PivotTable is shown with the following structure:

|             | Average Age=AVERAGE(Customer[age]) |
|-------------|------------------------------------|
| Average Age | 42,46128                           |

The status bar at the bottom indicates 'Customer | Product | Sales | Store | Calendar' and 'PivotTable: 1 of 25,000'.

Measure Average Age dibuat pada tabel Customer untuk menghitung rata-rata usia pelanggan yang terdapat pada dataset. Perhitungan dilakukan menggunakan fungsi AVERAGE() pada kolom age sehingga sistem dapat menghitung nilai rata-rata umur seluruh customer secara otomatis. Formula yang digunakan adalah AVERAGE(Customer[age]). Measure ini bekerja dengan menjumlahkan seluruh nilai usia pelanggan kemudian membaginya dengan jumlah customer yang tersedia pada data. Hasil dari measure bersifat dinamis, sehingga nilainya dapat berubah sesuai filter atau analisis yang diterapkan pada PivotTable maupun Data Model. Penggunaan measure Average Age membantu dalam memahami karakteristik pelanggan berdasarkan rata-rata usia customer yang melakukan transaksi pada dataset penjualan.

Insert Calculated Field

?

×

Name:

profit

▼

Modify

Formula:

= list\_price - cost\_price

Delete

Fields:

product\_id

category

color

size

season

supplier

cost\_price

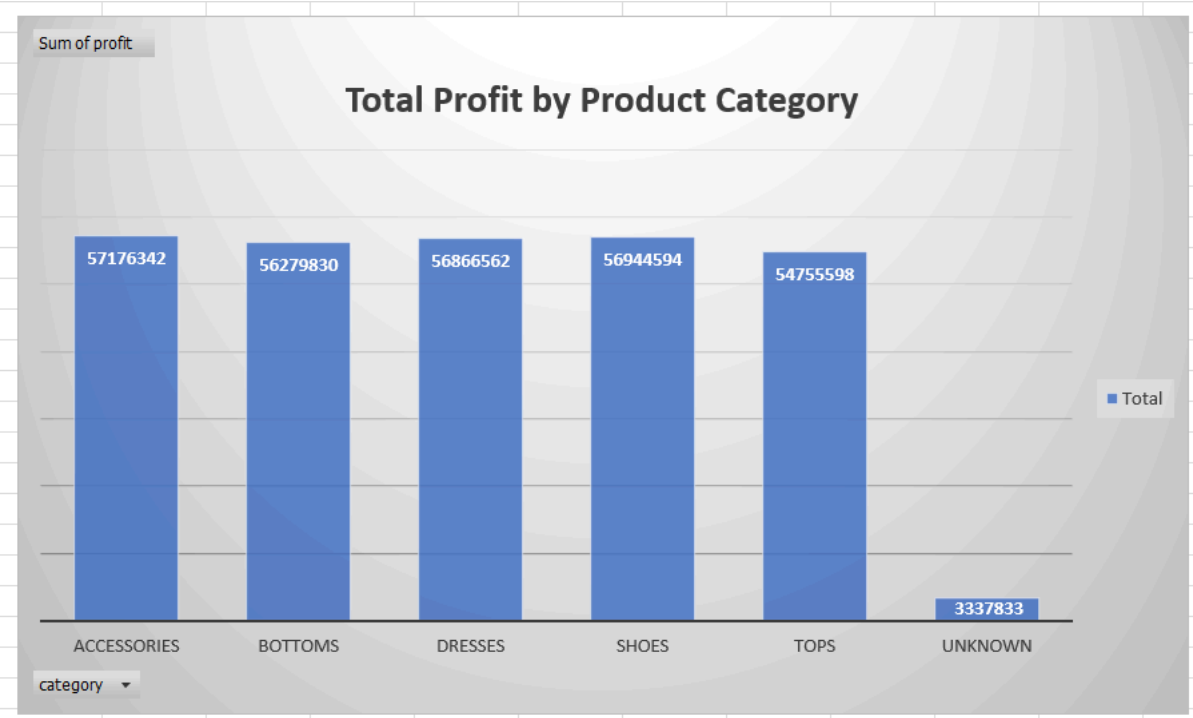
list\_price

Insert Field

OK

Close

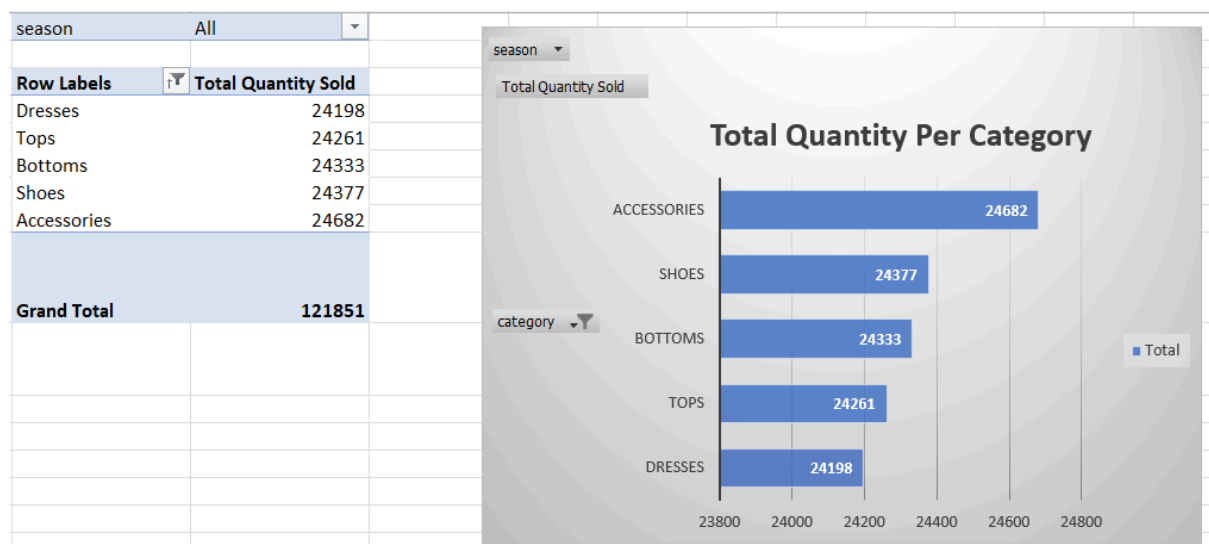
| Row Labels         | Sum of profit    |
|--------------------|------------------|
| Accessories        | 57176342         |
| Bottoms            | 56279830         |
| Dresses            | 56866562         |
| Shoes              | 56944594         |
| Tops               | 54755598         |
| Unknown            | 3337833          |
| <b>Grand Total</b> | <b>285360759</b> |



Pada analisis ini dibuat sebuah **Calculated Field** bernama **Profit** pada Pivot Table untuk menghitung keuntungan yang diperoleh dari setiap produk. Calculated Field tersebut menggunakan rumus **Profit = List Price – Cost Price**, di mana *List Price* merupakan harga jual produk dan *Cost Price* merupakan biaya perolehan produk. Dengan perhitungan ini, setiap produk memiliki nilai keuntungan yang kemudian dapat dijumlahkan dan dianalisis berdasarkan kategori produk. Penggunaan Calculated Field memungkinkan perhitungan dilakukan secara otomatis di dalam Pivot Table tanpa perlu menambahkan kolom baru pada dataset asli, sehingga proses analisis menjadi lebih efisien dan dinamis.

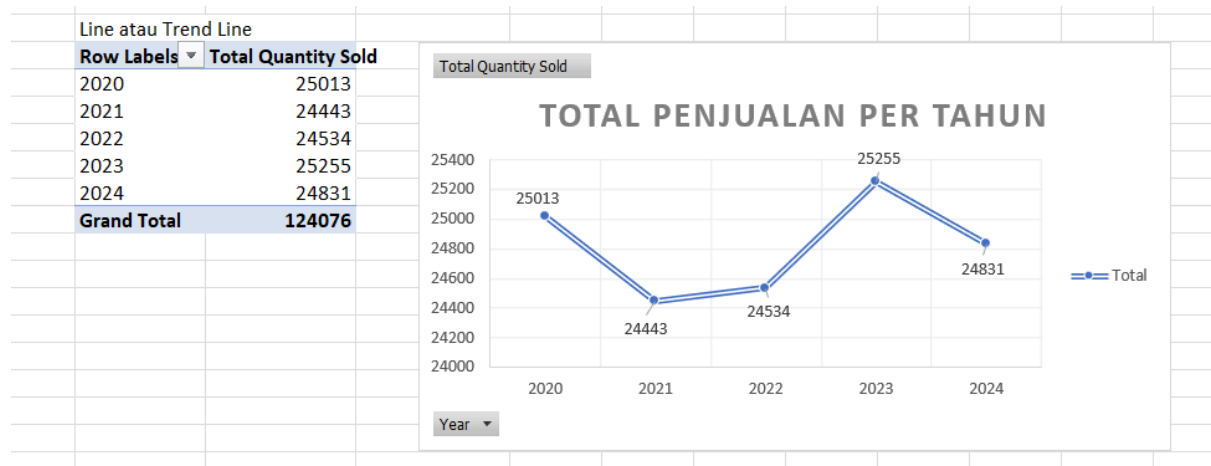
## 7.Mastering Charts

- Bar Chart



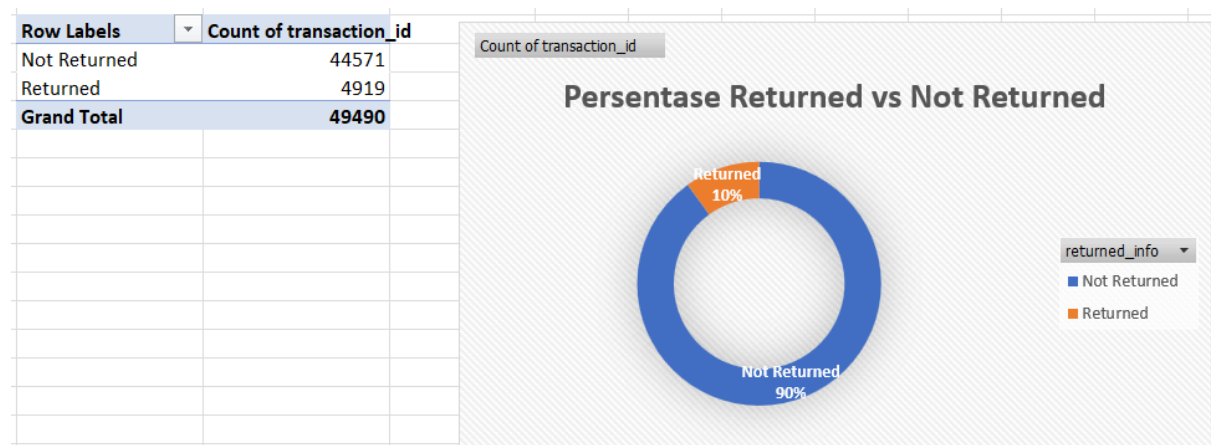
Bar Chart digunakan untuk membandingkan total quantity penjualan pada setiap kategori produk yang terdapat pada dataset. Grafik batang mempermudah visualisasi kategori dengan jumlah penjualan tertinggi dan terendah sehingga perbandingan antar kategori dapat dilihat dengan lebih jelas. Dengan menggunakan chart ini, analisis penjualan produk menjadi lebih mudah dipahami secara visual.

- Line atau Trend Line



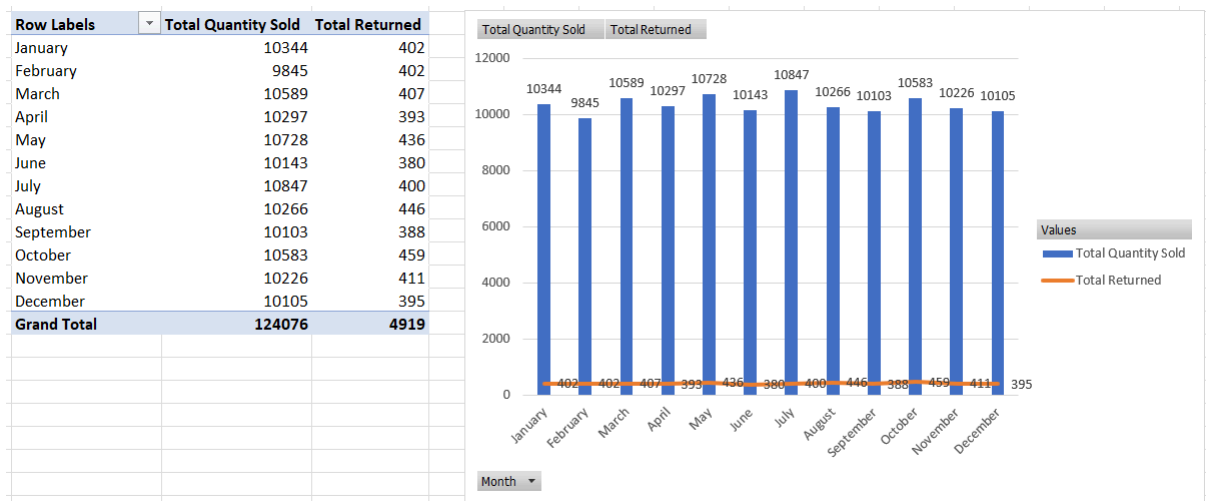
Line Chart digunakan untuk melihat tren penjualan berdasarkan periode waktu tahunan. Grafik garis membantu menunjukkan pola kenaikan maupun penurunan jumlah penjualan dari waktu ke waktu sehingga perkembangan penjualan dapat dianalisis dengan lebih jelas. Visualisasi ini mempermudah identifikasi bulan dengan penjualan tertinggi maupun terendah pada dataset.

- Doughnut Chart



Doughnut Chart digunakan untuk menunjukkan persentase transaksi returned dan not returned pada dataset penjualan. Grafik ini membantu memvisualisasikan proporsi data dalam bentuk persentase sehingga perbandingan antara produk yang direturn dan tidak direturn dapat terlihat dengan lebih mudah. Dari hasil visualisasi terlihat bahwa sebagian besar transaksi termasuk kategori not returned dibandingkan returned.

- Combination Chart



Combination Chart digunakan untuk menggabungkan dua jenis visualisasi dalam satu grafik, yaitu column chart untuk total quantity penjualan dan line chart untuk jumlah returned berdasarkan periode bulanan. Grafik batang menunjukkan total produk yang terjual setiap bulan, sedangkan grafik garis menunjukkan jumlah produk yang direturn pada bulan yang sama. Visualisasi ini membantu menganalisis hubungan antara peningkatan penjualan dengan tingkat returned sehingga pola transaksi dapat dipahami dengan lebih baik.

- Column Chart



Grafik **Total Profit by Product Category** menampilkan total keuntungan yang diperoleh dari setiap kategori produk berdasarkan calculated field **Profit**, yang dihitung menggunakan rumus **Profit = List Price – Cost Price**. Rumus ini digunakan untuk mengetahui selisih antara harga jual dan biaya perolehan produk sehingga dapat menggambarkan keuntungan yang

dihasilkan oleh setiap produk. Berdasarkan hasil visualisasi, kategori **Accessories** menghasilkan total profit tertinggi sebesar sekitar 57,18 juta, diikuti oleh kategori Dresses, Shoes, Tops, dan Bottoms yang memiliki nilai profit relatif serupa. Sementara itu, kategori **Unknown** menghasilkan profit yang jauh lebih rendah dibandingkan kategori lainnya. Analisis ini membantu perusahaan mengidentifikasi kategori produk yang memberikan kontribusi keuntungan terbesar sehingga dapat menjadi fokus dalam strategi penjualan dan pengelolaan produk.

- Creating Interactive Dashboard



Dashboard ini dibuat untuk memvisualisasikan dan menganalisis data penjualan perusahaan secara interaktif menggunakan Excel, Power Pivot, Pivot Table, Pivot Chart, Measure, Calculated Field, dan Cube Functions. Tujuan utama dashboard ini adalah membantu

pengguna memahami performa penjualan, profit, karakteristik pelanggan, serta tingkat pengembalian produk melalui satu halaman yang mudah dibaca dan dianalisis.

Pada bagian atas dashboard terdapat tiga Key Performance Indicators (KPI) yang ditampilkan menggunakan Cube Functions. KPI pertama adalah **Average Age** sebesar 42,46 tahun yang menunjukkan rata-rata usia pelanggan. KPI kedua adalah **Total Quantity Sold** sebesar 124.076 unit yang menunjukkan total produk yang berhasil terjual. KPI ketiga adalah **Total Returned** sebesar 4.919 transaksi yang menunjukkan jumlah produk yang dikembalikan oleh pelanggan. Ketiga KPI ini memberikan gambaran umum mengenai kondisi bisnis secara cepat sebelum pengguna melihat detail visualisasi lainnya.

Visualisasi pertama adalah **Total Quantity per Category** yang menunjukkan jumlah produk terjual pada setiap kategori produk. Hasil analisis menunjukkan bahwa kategori Accessories memiliki jumlah penjualan tertinggi, sedangkan kategori Dresses memiliki jumlah penjualan terendah. Grafik ini membantu perusahaan mengetahui kategori produk yang paling diminati pelanggan.

Visualisasi kedua adalah **Total Penjualan per Tahun** yang menampilkan tren penjualan dari tahun 2020 hingga 2024. Dari grafik terlihat bahwa penjualan mengalami penurunan pada tahun 2020, kemudian meningkat hingga mencapai puncaknya pada tahun 2023 sebelum sedikit menurun pada tahun 2024. Grafik ini membantu melihat perkembangan performa bisnis dari tahun ke tahun.

Visualisasi ketiga adalah **Persentase Returned vs Not Returned** yang membandingkan jumlah transaksi yang mengalami pengembalian produk dengan transaksi yang tidak mengalami pengembalian. Hasil analisis menunjukkan bahwa sekitar 90% transaksi tidak mengalami return, sedangkan hanya sekitar 10% transaksi yang direturn. Hal ini menunjukkan bahwa sebagian besar transaksi berjalan dengan baik dan tingkat pengembalian produk relatif rendah.

Visualisasi keempat adalah **Sales Trend dan Returned Trend per Month** yang menggunakan combination chart untuk membandingkan jumlah penjualan dan jumlah return pada setiap bulan. Grafik ini membantu mengidentifikasi apakah peningkatan penjualan pada periode tertentu diikuti oleh peningkatan jumlah produk yang dikembalikan oleh pelanggan.

Visualisasi terakhir adalah **Total Profit by Product Category** yang dibuat menggunakan Calculated Field bernama Profit dengan rumus **List Price – Cost Price**. Grafik ini menunjukkan total keuntungan yang dihasilkan oleh setiap kategori produk. Hasilnya menunjukkan bahwa kategori Accessories memberikan kontribusi profit terbesar, sedangkan kategori Unknown menghasilkan profit paling rendah. Informasi ini dapat digunakan untuk menentukan kategori produk yang paling menguntungkan bagi perusahaan.